

Trainings ▾

General Information

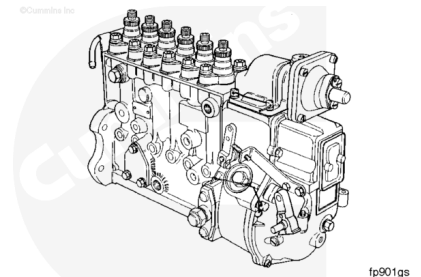
Fuel System Components and Flow - Distributor-Type Fuel Injection Pump

Most of the engines will be equipped with a cam actuated fuel transfer pump. Two types of fuel pumps are available for distributor-type fuel injection pumps: 1) A diaphragm-style fuel transfer pump and 2) a piston-style fuel transfer pump. The piston-style fuel transfer pump is offered as an option. Fuel flow begins as the fuel transfer pump pulls fuel from the supply tank through a mandatory prefilter. The fuel transfer pump supplies low-pressure fuel to the fuel filter head, through the filter, and then to the distributor injection pump.

The distributor pump builds the high injection pressures required for combustion and routes the fuel through individual high-pressure fuel lines to each injector.

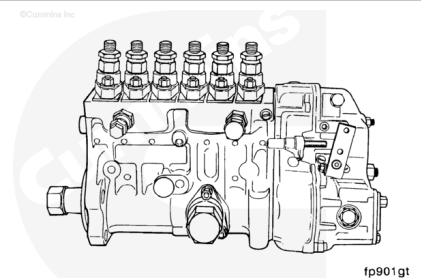
When the high-pressure fuel reaches the injector, the pressure lifts the needle valve against the spring tension to let the fuel enter the combustion chamber.

Any leakage past the needle valve enters the fuel drain manifold. The fuel drain manifold routes controlled venting from the distributor injection pump and leakage from the injectors back into the fuel tank.



Fuel System Components and Flow - In-Line Fuel Injection Pump

A cam-actuated piston-style fuel transfer pump provides positive fuel pressure to the fuel injection pump. Fuel flow begins as the fuel transfer pump pulls fuel from the supply tank. The system should have a prefilter or screen to remove larger contaminants from the fuel before reaching the fuel transfer pump. The fuel transfer pump supplies low-pressure fuel through the fuel filter head and filter, to the fuel injection pump.



The Bosch® in-line fuel injection pump is used on higher rated '91 and '94 B Series automotive engines and '96 marine engines. The Nippondenso EP-9 was used on pre-'96 higher horsepower marine B Series engines.

The Bosch® P3000 fuel injection pump is used on higher horsepower certified industrial engines and lower horsepower '91 to '93 automotive engines.

Bosch® "A"-type fuel injection pumps are used on lower horsepower certified industrial engines.

The fuel injection pump builds the high injection pressures required for combustion and routes the fuel through high-pressure fuel lines to each injector.

All B Series engines use closed-nozzle, hole-type injectors. When the high-pressure fuel reaches the injector, the pressure lifts the needle valve against the spring tension to allow fuel to enter the combustion chamber.

Any leakage past the needle valve enters the fuel drain manifold.

On engines equipped with the Bosch® in-line fuel injection pumps, the manifold routes leakage from the injectors to the inlet side of the fuel filter. On engines equipped with the Nippondenso EP-9 fuel injection pump, the manifold routes leakage from the injectors to the supply tank.